Ryan Cross

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Security Clearance: Top Secret (TS) (In Progress)

Honors graduate pursuing M.S. in Aerospace Engineering with 3+ years of professional and leadership experience. Committed to continuous learning to make a positive impact on the planet and drive a more sustainable future.

Professional Skills

4+ years in software engineering | **MATLAB**, **Python**, **C++**, **GitHub**, **Linux**, and **OpenC3 COSMOS**
2+ years in computational analysis | **SimScale**, **OpenFOAM**, **Simcenter STAR-CCM+**, **ANSYS Fluent** and **ParaView**
2+ years in design & manufacturing | **SOLIDWORKS**, **Siemens NX**, **OnShape**, **OpenVSP**, and **3D printing**
1+ years in systems engineering | **SysML**, **MBSE**, **Cameo Enterprise Architecture**, and **Velocity Template Language**

Work Experience

SIERRA SPACE | ***Systems Engineering Intern*** *May 2026 - Present*
$8 Billion aerospace leader building next-generation spacecraft and space infrastructure

- Refine and manage 300+ spacecraft requirements across 6 levels of development through SysML-based MBSE
- Automate technical document generation using Velocity Template Language (VTL) to enable model consistency
- Develop OpenC3 COSMOS thermal response protocols to safeguard $900K in flight-representative hardware

KASHMIR WORLD FOUNDATION | ***Aerodynamics Intern*** *Dec 2025 - Sep 2026*
Nonprofit team advancing wildlife conservation through blended-wing body UAS technology

- Conducted CFD analysis from 0-20 AoA, using results to drive design changes and identify elevon trim angles
- Resolved CAD model in OnShape and refined mesh to achieve $y+ < 1$ for optimized solution convergence
- Documented setup, strategy, and modeling in a published technical paper to justify results

KAIROS POWER | ***Test Engineering Intern*** *May - Aug 2025*
Nuclear energy company developing advanced molten-salt reactor technology

- Designed and built a 6-ft reactor coolant test section to fabricate and collect experimental void fraction data
- Computed theoretical void fraction predictions using established two-phase flow models and analyses
- Created a multidimensional optimization function in MATLAB to plot prediction error in various 3D spaces

Projects

CU DESIGN BUILD FLY | ***Lead Propulsion Engineer*** *Apr 2024 - May 2026*
Student team designing, building, and flying mission-optimized aircraft for AIAA competition

- Designed critical aircraft components using CAD and produced 4 prototype stages with CAM techniques
- Collaborated on mission proposal by applying technical writing methods, placing top 25% after AIAA official review

CU BOULDER SENIOR CAPSTONE | ***Design and Test Team*** *Aug 2025 - May 2026*
Goal: Improve Boeing 737 MAX-8 efficiency by 50% across a 40-member team

- Assembled ground station for prototype demonstrator, executing design process in a project PDR and CDR

Education

University of Colorado Boulder *Aug 2025 - May 2027*
M.S. Aerospace Engineering - Fluids, Structures, Materials *GPA: 4.0/4.0*

University of Colorado Boulder *Aug 2022 - May 2026*
B.S. Aerospace Engineering *GPA: 3.7/4.0*